

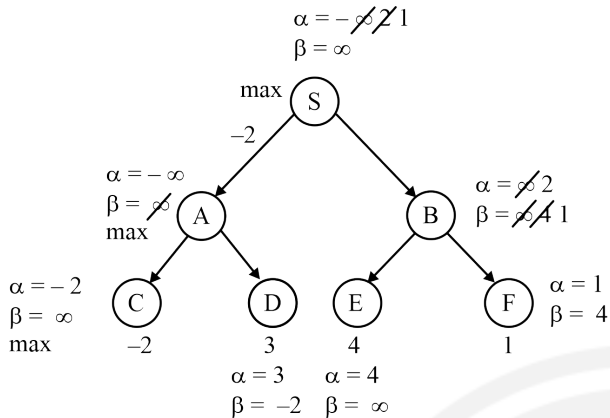
DS&AI

Artificial Intelligence

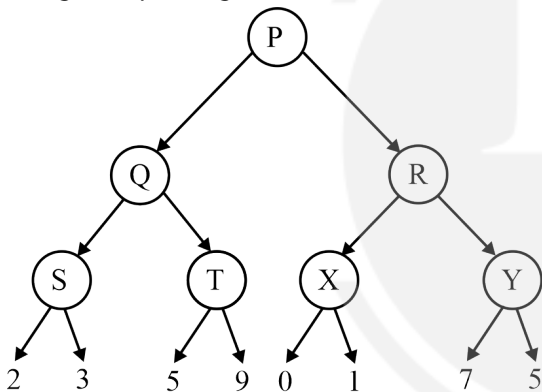
DPP: 1

Adversarial Search

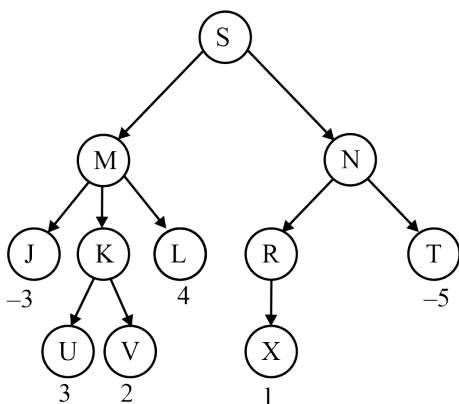
Q1



Q2 Count the number of nodes that will be pruned using a α - β pruning

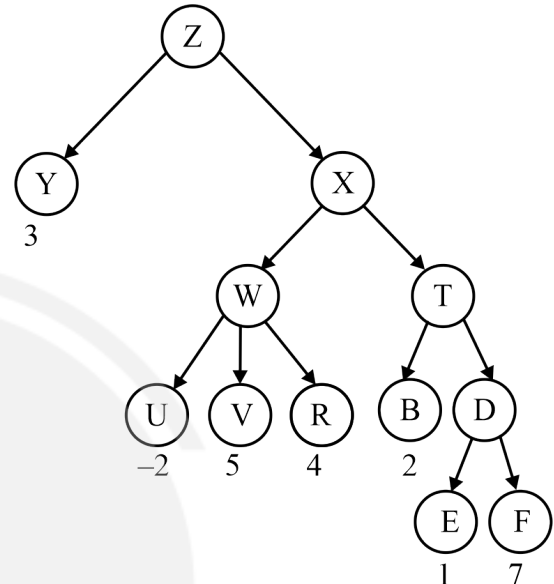


Q3 Consider the tree green below. The root node S is the max player. What will be the best score for this root node S.

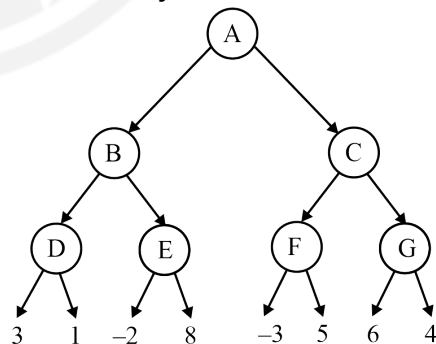


Q4

The best score possible for the root node which is a minimum player is ____.



Q5 Consider the following tree. Let X denote the best score of the root node when it acts as the 'max' player. Let Y denote the best score of the root node when it acts as the 'min' player. The value of $x + y$ is ____.



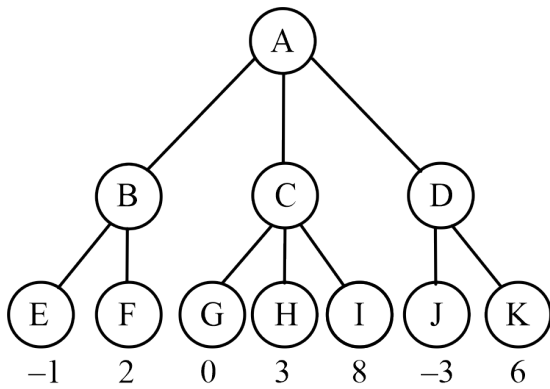
Q6 Find the best result for the root where root acts as the max player:



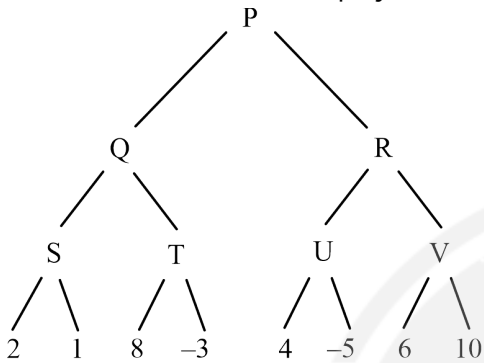
Android App

iOS App

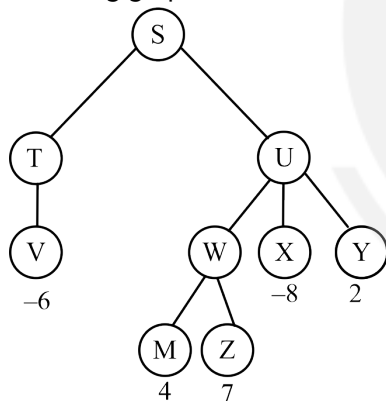
PW Website



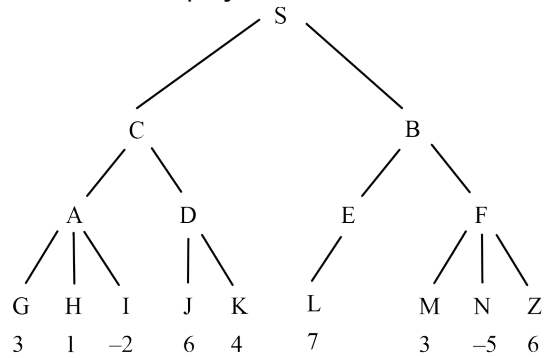
Q7 Find the best result for min player:



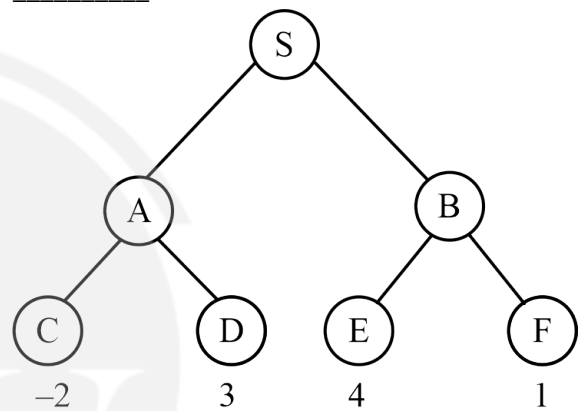
Q8 Find the best score for S (max player) in the following graph:



Q9 What is the best result for node C if the root node is a max player?



Q10 The number of nodes that will be pruned using alpha-beta pruning in the following graph is _____.



Answer Key

Q1 0
Q2 4
Q3 3
Q4 -2
Q5 6

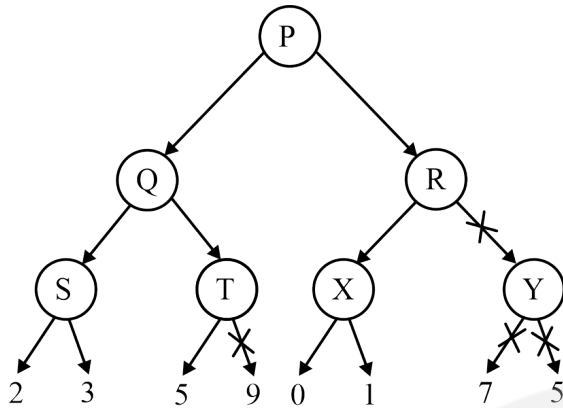
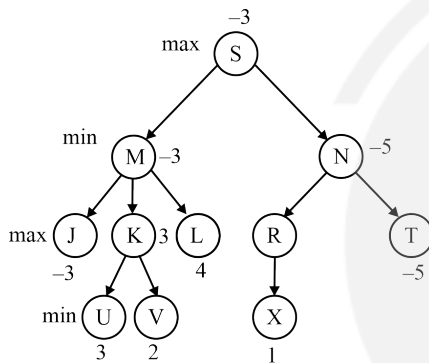
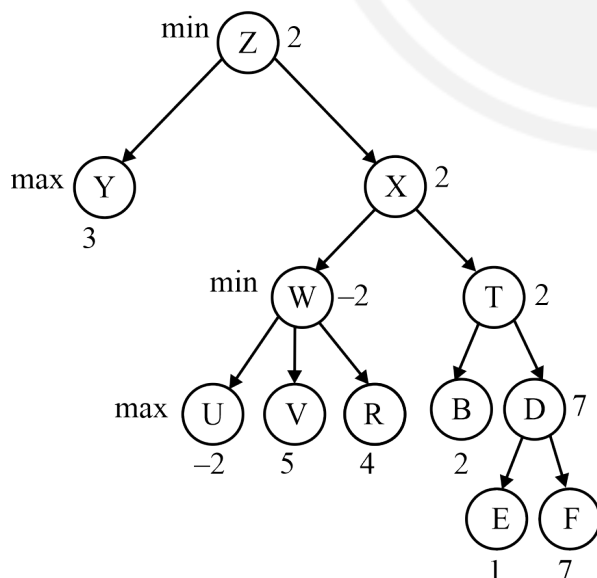
Q6 0
Q7 -1
Q8 -6
Q9 -3
Q10 0

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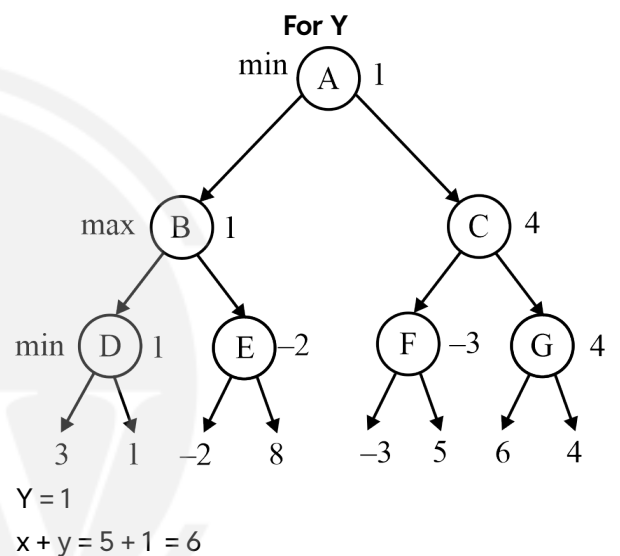
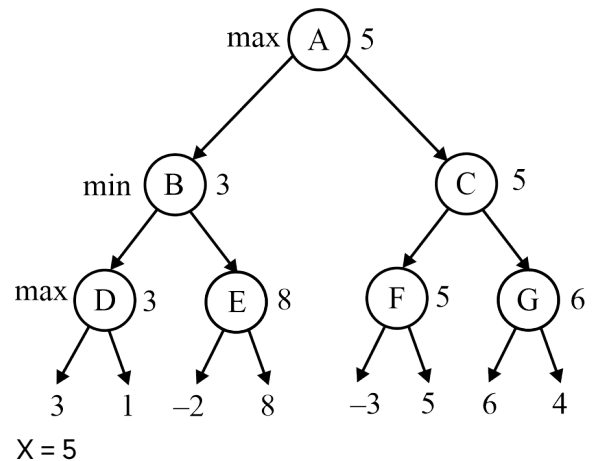
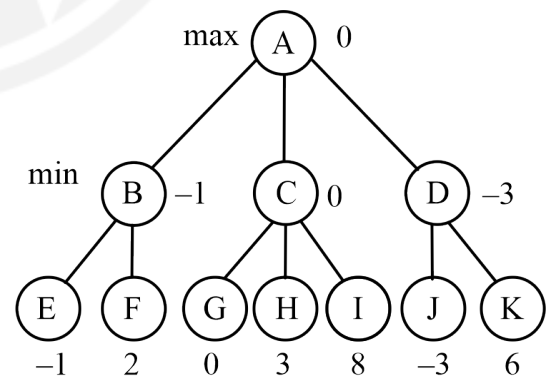
Hints & Solutions

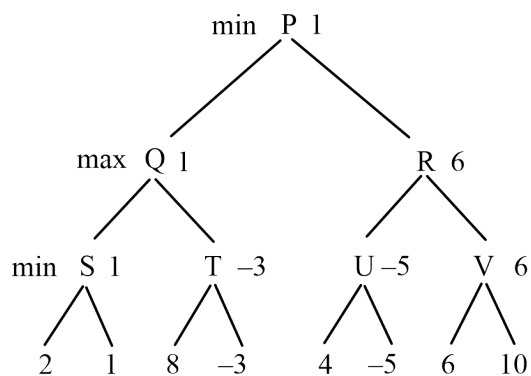
Q1 Text Solution:

No node will be pruned.

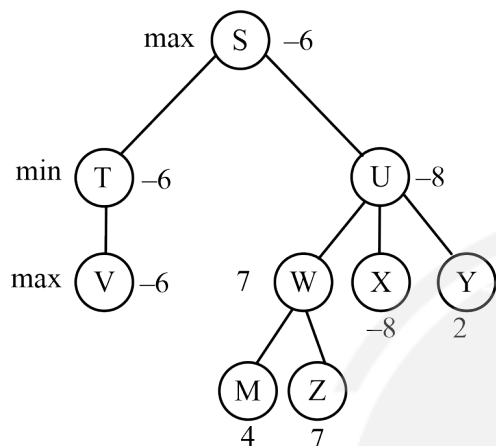
Q2 Text Solution:

Q3 Text Solution:

Q4 Text Solution:

Q5 Text Solution:

For x

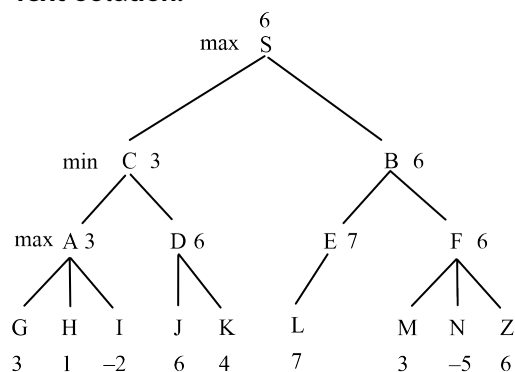

Q6 Text Solution:

Q7 Text Solution:

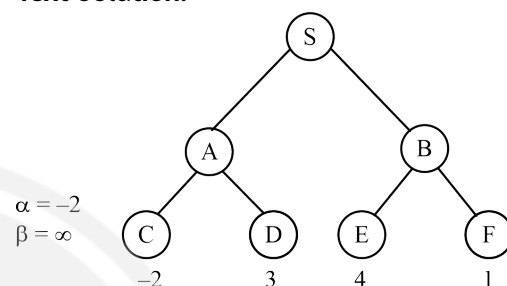
Q8 Text Solution:



Q9 Text Solution:



Q10 Text Solution:



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